



TrashUQ

Edge AI + Federated Learning Platform

Aleix Bertran Andreu · Aleix Rosinach
Bru · Pol Llinàs

May 2026 · Real-Time Monitoring & Scaling Validation

Project Overview

TrashUQ is an end-to-end platform for intelligent waste monitoring.

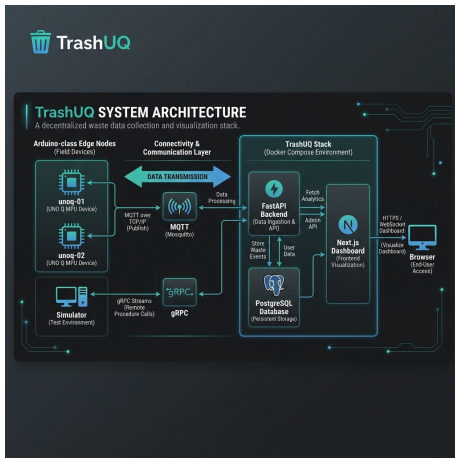
- **Decentralized Training:** Federated Learning at the Edge.
- **Real-Time Pipeline:** MQTT telemetry & WebSocket dashboard.
- **Robust Persistence:** PostgreSQL storage of every event.
- **Validated Hardware:** Deployed on Arduino-class MPU nodes.

Core Stack

- **Backend:** FastAPI & gRPC
- **Edge:** TFLite & Python
- **Dashboard:** Next.js
- **Broker:** Mosquitto



System Architecture



Three-Stage Pipeline: Edge Devices → Connectivity → Deployable Stack

Device Layer

- **Hardware:** UNO Q MPU (Arduino-class).
- **Runtime:** TFLite-based material classification.
- **Simulator:** Full-fidelity reproducible demo mode.

Classification Path

- Cardboard
- Glass
- Paper
- Plastic

Telemetry: Published via MQTT topics.

The Communication Backbone

MQTT Telemetry

- Real-time status, metrics, and classification streams.
- Low-bandwidth, high-reliability pub/sub.

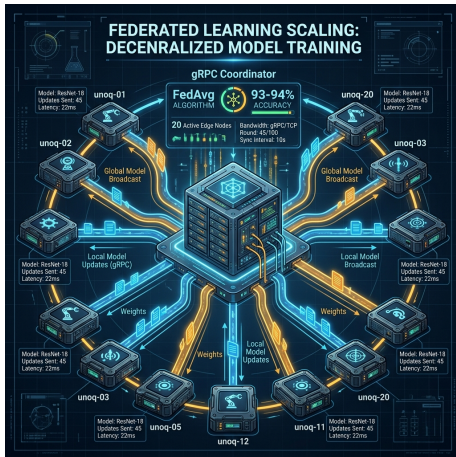
gRPC Coordination

- High-performance binary RPC for model weight transfer.
- Concurrency-safe: Rejects stale updates during rounds.

Data Flow

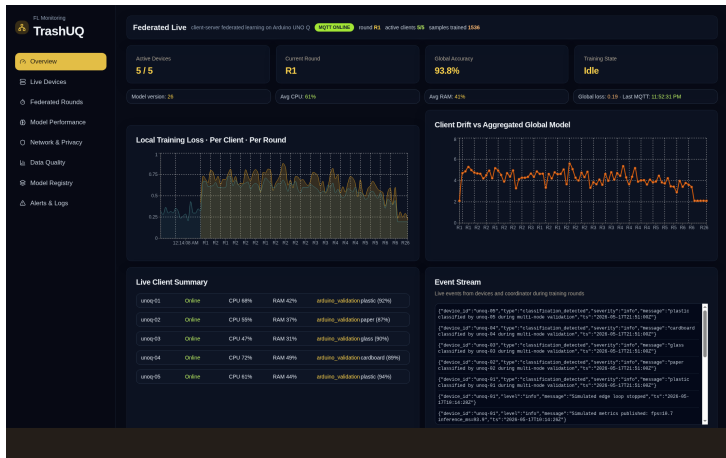
arduino/+/# → Backend →
PostgreSQL → Dashboard

Federated Learning Scaling



FedAvg Validation: Scalability study across 2, 5, 10, and 20 clients.

Next.js Live Dashboard



Bootstrap + MQTT WebSockets: Real-time charts and device summaries.



Part A: Deployed Validation

- **Infrastructure:** 2 nodes (unoq-01, unoq-02) on bepes-server.
- **Persistence:** 282 MQTT messages stored with full fidelity.
- **Monotonicity:** Advanced `model_version` from 0 to 26 over 26 rounds.
- **Reliability:** 0% invalid payloads, 0% dropped messages.

Node	Messages	Mean Accuracy
unoq-01	147	92.1%
unoq-02	135	87.1%

Part B: Scaling Performance

- **Experiment:** FedAvg scaling study (2 to 20 clients).
- **Condition:** Non-IID Dirichlet partitioning ($\alpha = 0.3$).
- **Convergence:** Stable convergence with **93-94% final accuracy**.
- **Efficiency:** predictable linear growth in communication cost.

Conclusion: System maintains high performance at scale under heterogeneous data.

TrashUQ Success

- Full edge-to-cloud integration.
- Validated FL coordinator.
- Real-time dashboard.

The Team

- Aleix Bertran Andreu
- Aleix Rosinach
- Bru
- Pol Llinàs

Thank You!